

Wei Fang

PHD CANDIDATE AT MIT CSAIL

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Research Interests

LLMs, Tool-augmented Generation, LLM Reasoning, Information Retrieval

Education

Massachusetts Institute of Technology

Cambridge, MA, USA

PH.D. CANDIDATE, ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

2025 (Expected)

- Advisor: Dr. James Glass
- Thesis: Generalizing Augmented Large Language Models.

M.S., ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

2021

- Advisor: Dr. James Glass
- Thesis: On End-to-end Automatic Fact-checking Systems

National Taiwan University

Taipei, Taiwan

B.S.E., ELECTRICAL ENGINEERING

2018

Experience

MIT CSAIL Spoken Language Systems Group

Cambridge, MA, USA

RESEARCH ASSISTANT, SUPERVISED BY DR. JAMES GLASS

Sep. 2018 - Present

- **Tool-augmented Generation and Reasoning** Tool instruction optimization to improve LLMs' tool-use [ACL25], and planning with tools [WIP].
- **Information retrieval** Passage retrieval for open-domain QA — passage re-ranking [EACL24] and query expansion [ACL23].
- **Factuality & verification** Factual and fair language generation [ASRU23W]; Automatic fact-checking system [NAACL19]; Stance detection with multi-task transfer learning [EMNLP19W].
- **Conversational QA** Question generation for conversational question answering [ACL23W].
- **Information extraction** Cross-document temporal relation extraction by anchoring on temporal events [Preprint].
- **Speech synthesis** End-to-end speech synthesis with pre-trained natural language representations [Preprint].

IBM Research MIT-IBM Watson AI Lab

Cambridge, MA, USA

RESEARCH INTERN

Jun. 2024 - Aug. 2024

- Tool instruction optimization to improve LLMs' tool-use [ACL25].

Apple Inc. Natural Language Processing Team

Cupertino, CA, USA

RESEARCH INTERN

Jul. 2018 - Aug. 2018

- Generative modeling for the QuickPath project.

RESEARCH INTERN

Jun. 2017 - Sep. 2017

- Input representations for neural models to improve user keyboard experience.

NTU & Academia Sinica Vision & Learning Lab

Taipei, Taiwan

RESEARCH ASSISTANT, SUPERVISED BY PROF. YU-CHIANG FRANK WANG

Jul. 2016 - Jan. 2018

- Incorporate knowledge graphs for multi-label zero-shot image recognition [CVPR18].

National Taiwan University (NTU) Speech Lab

Taipei, Taiwan

RESEARCH ASSISTANT, SUPERVISED BY PROF. LIN-SHAN LEE & PROF. HUNG-YI LEE

Sep. 2015 - Jun. 2017

- Hierarchical attention-based model for the TOEFL Listening Comprehension Test [SLT16] [TASLP19].
- Unsupervised audio word embeddings and demo.

Publications

[1] Wei Fang, Yang Zhang, Kaizhi Qian, James Glass, Yada Zhu. PLAY2PROMPT: Zero-shot Tool Instruction Optimization for LLM Agents via Tool Play. *Findings of the Association for Computational Linguistics: ACL*, 2025.

[2] Wei Fang, Yung-Sung Chuang, James Glass. Joint Inference of Retrieval and Generation for Passage Re-ranking. *Findings of the Association for Computational Linguistics: EACL*, 2024.

[3] Tianhua Zhang, Hongyin Luo, Yung-Sung Chuang, Wei Fang, Luc Gaitskell, Thomas Hartvigsen, Xixin Wu, Danny Fox, Helen Meng, James Glass. Interpretable Unified Language Checking. *Symposium for Celebrating 40 Years of Bayesian Learning in Speech and Language Processing and Beyond at ASRU*, 2023.

[4] Yung-Sung Chuang, Wei Fang, Shang-Wen Li, Wen-tau Yih, James Glass. Expand, Rerank, and Retrieve: Query Reranking for Open-domain Question Answering. *Findings of the Association for Computational Linguistics: ACL*, 2023.

[5] Tianhua Zhang, Liping Tang, Wei Fang, Hongyin Luo, Xixin Wu, Helen Meng, James Glass. ConvRGX: Recognition, Generation, and Extraction for Self-trained Conversational Question Answering. *DialDoc Workshop on Document-grounded Dialogue and Conversational Question Answering at ACL*, 2023.

[6] Wei Fang and James Glass. Cross-Document Temporal Relation Extraction with Temporal Anchoring Events. *Preprint*, 2021.

[7] Wei Fang. On End-to-end Automatic Fact-checking Systems. *S.M. Thesis*, 2021.

[8] Wei Fang, Moin Nadeem, Mitra Mohtarami, James Glass. Neural Multi-task Learning for Stance Prediction. *EMNLP-IJCNLP Workshop on Fact Extraction and VERification*, 2019.

[9] Wei Fang, Yu-An Chung, James Glass. Towards Transfer Learning for End-to-end Speech Synthesis from Deep Pre-trained Language Models. *arXiv Preprint*, 2019.

[10] Chia-Hsuan Lee, Hung-Yi Lee, Szu-Lin Wu, Chi-Liang Liu, Wei Fang, Jui-Yang Hsu, Bo-Hsiang Tseng. Machine Comprehension of Spoken Content: TOEFL Listening Test and Spoken SQuAD. *IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, 2019.

[11] Moin Nadeem, Wei Fang, Brian Xu, Mitra Mohtarami, James Glass. FAKTA: An Automatic End-to-End Fact Checking System. *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019.

[12] Chung-Wei Lee[†], Wei Fang[†], Chih-Kuan Yeh, Yu-Chiang Frank Wang. Multi-Label Zero-Shot Learning with Structured Knowledge Graphs. *IEEE Conference on Computer Vision & Pattern Recognition (CVPR)*, 2018.

[13] Wei Fang[†], Jui-Yang Hsu[†], Hung-Yi Lee, Lin-Shan Lee. Hierarchical Attention Model for Improved Comprehension of Spoken Content. *IEEE Workshop on Spoken Language Technology (SLT)*, 2016.

[†] indicates equal contribution

Professional Activity

Reviewer	ACL Rolling Review (2021-), ACL (2021-), EMNLP (2021-), NeurIPS (2024-), ICML (2025-)
Affiliations	Sigma Xi, the Scientific Research Honor Society (2025-)

Teaching

6.864 Advanced Natural Language Processing , Teaching Assistant	Spring 2021
EE5177/EE5184 Machine Learning , Teaching Assistant	Fall 2016, Spring 2017
CSIE5610 Data Analytics & Modeling , Teaching Assistant	Fall 2016
EE2011 Signals and Systems , Teaching Assistant	Spring 2016

Honors

Government Scholarship to Study Abroad , Ministry of Education, Taiwan	May 2021
Student Travel Bursary , Amazon	Oct. 2019
EECS Great Educators Fellowship , MIT EECS	Feb. 2018
Dean's List , NTU Electrical Engineering	2014-2016
Conference Grant , Foundation for the Advancement of Outstanding Scholarship	Dec. 2016